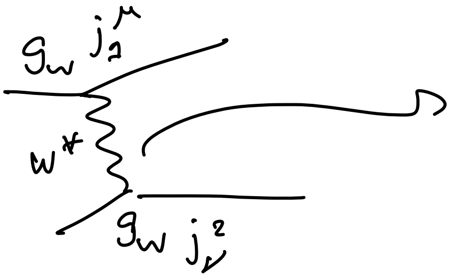
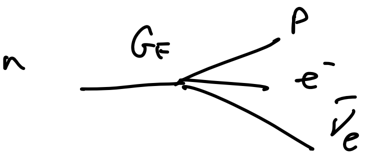
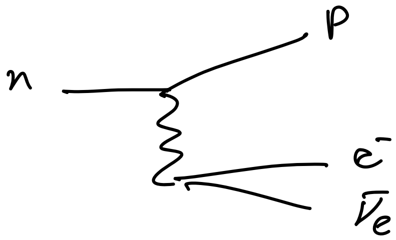
Weak Interaction



$$\frac{1}{q^2 - m_w^2}\left(g^{\mu\nu} - q^\mu q^\nu / m_w^2\right)$$

$$\mathcal{M} \sim \frac{g_w^2}{q^2 - m_w^2}\left(g^{\mu\nu} - q^\mu q^\nu / m_w^2\right) j_\mu^1\, j_\nu^2$$





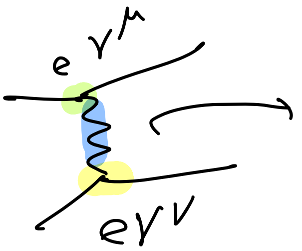
$$\mathcal{M} = G_F\, \underline{J}_{e\nu}\, \underline{J}_{pn}$$

$$\mathcal{M}_{weak} = \frac{g_w^2}{q^2 - m_w^2}\, \underline{J}_{np} \cdot \underline{J}_{e\nu}$$

Exp evidence: $\mathbb{C}$ violated in weak interactions.

$\mathbb{P}$ violated

$\Rightarrow$ affect $\underline{J}_{e\nu}$, $\underline{J}_{np}$



$$\frac{g^{\mu\nu}}{q^2}$$

$$\mathcal{M}_{QED} \sim \frac{e^2}{4\pi}\left(\bar\psi \gamma^\mu \psi\right) \frac{g_{\mu\nu}}{q^2} \left(\bar\psi \gamma^\nu \psi\right)$$

$$= \alpha\, \underline{J}_1 \cdot \underline{J}_2$$

QED

$\underline{J} = \left(\bar\psi \gamma^0 \psi, \bar\psi \gamma^i \psi\right)$ transfors as a vector under $\mathbb{P}$

$\underline{J} = (J^0, \vec{J})$

$\mathbb{P}\, \underline{J} = (J^0, -\vec{J})$

γ^i : $i = 1, 2, 3$ Dirac matrices

$$\mathbb{P}\, \underline{J}_T = \mathbb{P}\left(\bar\psi' \gamma^0 \psi, \bar\psi \gamma^i \psi\right) = \left(\bar\psi \gamma^0 \psi, -\bar\psi \gamma^i \psi\right)$$

$$\mathcal{M}_{QED} = \frac{\alpha}{q^2}\left(j_1^0\, j_2^0 - \vec{j}_1 \cdot \vec{J}_2\right)$$

$\mathbb{P}\ \mathcal{M}_{QED} = \frac{\alpha}{q^2}\left(j_1^0 j_2^0 - (-\vec{j_1})\cdot(-\vec{j_2})\right) = \mathcal{M}_{QED}$

QED is invariant under $\mathbb{P}$

$\Rightarrow$ weak interaction $j_\mu^{weak} \neq \gamma_\mu$ because $\cancel{\mathbb{P}}$ in weak interactions.

what are possible forms for J_μ^{weak} ?

$\mathbb{P}\ \vec{V} \rightarrow -\vec{V}$ $\qquad \bar{\psi}\gamma^\mu\psi$

$\mathbb{P}\ S \rightarrow S$ $\qquad \bar{\psi}\psi$

$\mathbb{P}\ \vec{A} \rightarrow \vec{A}$ $\qquad$ pseudo-vector axial vector $\qquad \bar{\psi}\gamma^\mu\gamma^5\psi$

$\mathbb{P}\ ps \rightarrow -ps$ $\qquad$ pseudo-scaler $\qquad \bar{\psi}\gamma^5\psi$

$$\underline{j}_{weak} = \bar{\psi}\left(c_V\gamma^\mu + C_A\gamma^5\gamma^\mu + C_T\sigma^{\mu\nu}\right)\psi$$

$$\sigma^{\mu\nu} = \frac{i}{2}[\gamma^\mu,\gamma^\nu]$$

most generic form. $\qquad$ tensor bilinear

c_V, C_A, C_T: constants to be determined experimentally.

$$\mathcal{M}_{weak} = \frac{g_W^2}{q^2 - m_W^2}\ (\sim)\,c_V^2 + (\sim)\,C_A^2 + (\sim)\,c_V C_A + (\sim)\,C_T^2 + (\sim)\,c_V C_T \ldots$$

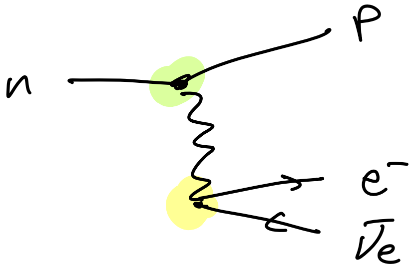
From experimental evidence

$$\underline{j}_{weak} = \bar{\psi}\gamma^\mu(1-\gamma^5)\psi$$
$$= \bar{\psi}\gamma^\mu\psi - \bar{\psi}\gamma^\mu\gamma^5\psi = V - A$$

charged weak current for fundamental fermions.

V–A theory $\qquad$ (for leptons and quarks (almost))

For hadrons



$\bar{\psi}_p \gamma^\mu (C_V + C_A \gamma^5) \psi_n$

C_V, C_A: to be measured exp.

(depends on hadrons)

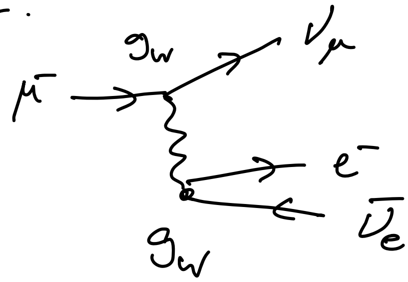
$\bar{\psi}_e \gamma^\mu (1-\gamma^5) \psi_\nu$

Experimental verification of V–A theory

– $n \to p e^- \bar{\nu}_e$ β^- decay.

– $\mu^- \to e^- \bar{\nu}_e \nu_\mu$

– $\pi^- \to e^- \bar{\nu}_e / \mu^- \bar{\nu}_\mu$



1) verify V–A theory prediction.

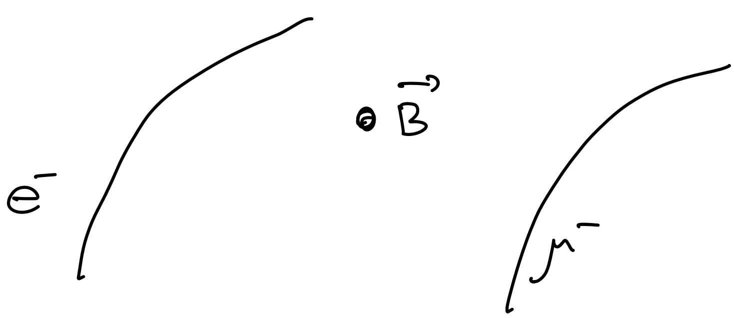
2) $g_W^e = g_W^\mu = g_W^\tau = g_W^q$?
is weak coupling constant same for all fermions?

QED

$e_{elec} = e_{muon}$?

from bending in the magnetic field

$p = 0.3\, z \times B \times R$

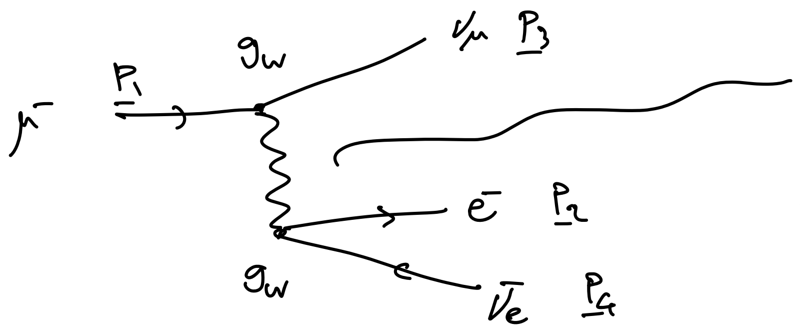


exp. evidence for $e_\mu \equiv e_e \equiv e_\tau$

to be tested also for g_W^e, g_W^μ, g_W^τ

Muon Decay

Griffiths 9.2 (Goldhaber ex 6.7)



$$\frac{g^{\mu\nu} - q^\mu q^\nu / m_w^2}{q^2 - m_w^2}$$

$|q| = ? \sim Q = m_\mu - m_e - m_\nu - m_{\bar{\nu}}$

$\mu^- \to e^- \bar{\nu}_e \nu_\mu$

$q^2 \simeq (100 \text{ MeV})^2 << m_w^2$

$$\mathcal{M} = \frac{g_w^2}{\cancel{q^2} - m_w^2}[\bar{u}^{\nu_\mu}(p_3)\gamma^\mu(1-\gamma^5)u^{\mu^-}(p_1)][\bar{u}(p_4)\gamma_\mu(1-\gamma^5)u(p_2)]$$

$\Gamma(\mu^- \to e^- \nu \bar{\nu}) \propto \langle|\mathcal{M}|^2\rangle \times$ (phase space)

$$\langle|\mathcal{M}|^2\rangle = \underset{\substack{\text{spins}\\ \text{in final}\\ \text{state}}}{\sum_i} \quad \underset{\substack{\text{average}\\ \text{spin}\\ \text{initial state.}}}{\sum} |\mathcal{M}|^2$$

Init. state: μ $\xrightarrow{\Leftarrow}$, μ $\xrightarrow{\Rightarrow}$ 2 states.

Final state: ν_μ $\xrightarrow{\Leftarrow}$, $\bar{\nu}_e$ $\xrightarrow{\Rightarrow}$

$\xrightarrow{\Rightarrow}$ $\xrightarrow{\Leftarrow}$ 2 states.

phase space: $$\rho = \frac{d^3 p_e}{(2\pi)^3}\frac{d^3 p_{\nu_\mu}}{(2\pi)^3}\frac{d^3 p_{\bar{\nu}_e}}{(2\pi)^3}\delta(\underline{p}_\mu - \underline{p}_e - \underline{p}_{\bar{\nu}_e} - \underline{p}_{\nu_\mu})$$

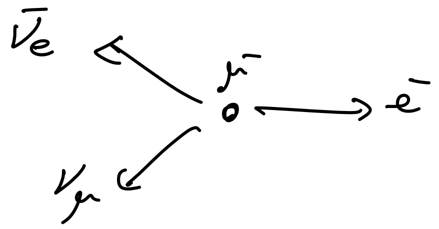
$\Gamma \propto |\mathcal{M}|^2 \times \rho$(phase space)

$$\mathcal{M} \sim \frac{g_w^2}{m_w^2} = G_F \Rightarrow |\mathcal{M}|^2 \sim G_F^2$$

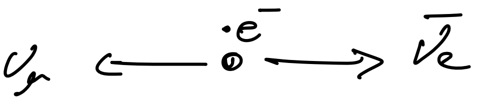
$[\mathcal{M}] = [E]^{-2}$ $\qquad [G_F^2] = [E]^{-4}$

$[\Gamma] = [E] \Rightarrow [\rho] = [E]^5$

In μ reference frame



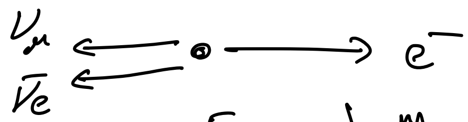
Min energy for e^-



$E_e = m_e \quad T_e = 0.$

$E_\nu = E_{\bar{\nu}} = \frac{1}{2} m_\mu.$

Max energy for e^-



$E_e = \frac{1}{2} m_\mu.$

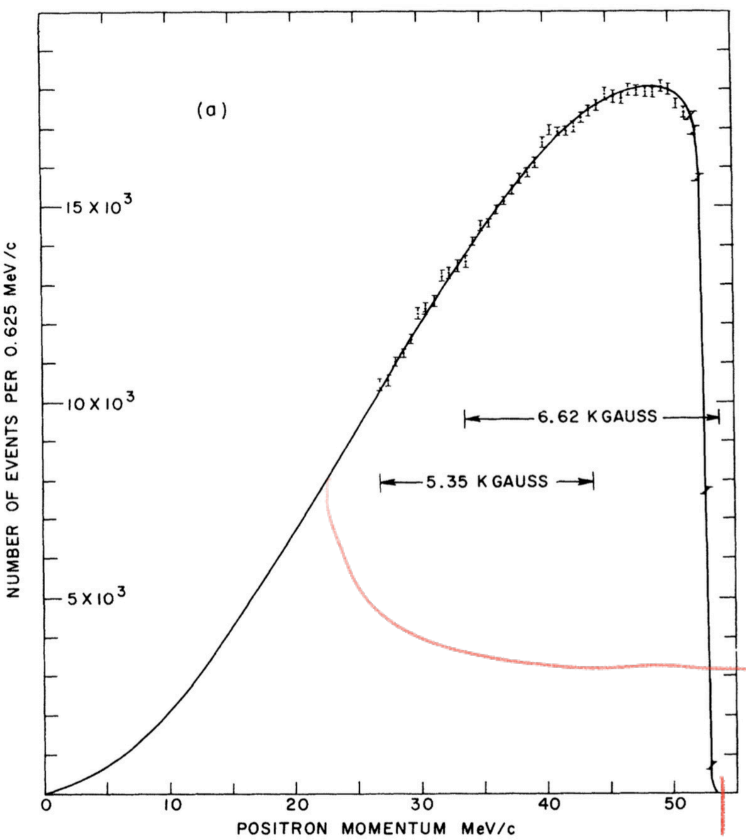
$E_e \in [m_e, \frac{1}{2} m_\mu]$

$\Gamma \propto G_F^2 m_\mu^5$ From dimensional analysis.

From full calculation

$$\Gamma_\mu = \frac{1}{192\pi^3} G_F^2 m_\mu^5$$

measure $\frac{d\Gamma_\mu}{dE_e}$ $\quad \frac{\#\ \mu \text{ decays}}{\text{en. of } e^- \text{ in decay}}.$



- produce many muons.
- count # decays.
- measure energy/mom of e^-/e^+

bin width = 0.625 MeV

V-A prediction